SecureChit - Technology Stack & Viva Guide

Full Stack Architecture & Explanations in Thanglish

■ 30-Second Quick Pitch (Viva Start-la Solla Vendiyathu):

*"Sir/Madam, namma project **Decoupled 4-Tier Architecture**-la develop pannirukkom.*
*• **Frontend:** React 18 with Vite & Tailwind CSS (for modern & responsive UI)*
*• **Backend:** Python FastAPI with Uvicorn (for high-speed async REST APIs)*
*• **Database:** PostgreSQL / SQLite with SQLAlchemy ORM & Alembic migrations*
*• **Security:** JWT stateless tokens + Bcrypt password encryption*
*• **Services:** ReportLab PDF invoices, UPI QR generation & SMTP Email alerts."*

1. Frontend (Client Tier)

• **React 18:** Component-based SPA (Single Page Application). Reusable components & fast state updates with Virtual DOM.
• **Vite 5 (Bundler):** Webpack / CRA vida romba fast-ah run aagum. Instant Hot Module Replacement (HMR) & lightning build time.
• **Tailwind CSS:** Utility-first CSS framework. Consistent theme, zero CSS conflicts & fast responsive mobile design.
• **React Router DOM v7:** Client-side routing. Admin routes & Member routes separate panni role-based security guard pannudhu.
• **Axios:** Backend API kooda communicate panna. Request interceptor moolama JWT token auto-ah attach aagum.
• **UI Libraries:** Lucide React (clean icons), React Hot Toast (instant toast alerts), React Custom Roulette & Canvas Confetti (auction draw animation).

2. Backend (API & Business Logic Tier)

• **Python FastAPI:** High-performance asynchronous framework (async/await). Django/Flask vida romba lightweight & fast.
• **Swagger Documentation:** Automatic interactive API documentation generate aagum (/docs URL moolama test pannalam).
• **Uvicorn:** FastAPI-kku ASGI web server. High concurrent connections smooth-ah handle pannum.
• **Pydantic v2:** Strict data validation. Request body inputs types & constraints auto-check panni invalid data thadukkum.
• **Authentication (JWT):** python-jose use panni stateless authentication & Role-Based Access Control (RBAC: Admin vs Member).
• **Password Security:** passlib[bcrypt] vachu plain text passwords salt panni hash aagi DB-la store aagum.
• **SlowAPI:** Rate limiting implement panni brute-force attacks and DDOS traffic prevent pannum.

3. Database Tier (Persistence & Migrations)

• **PostgreSQL / SQLite:** Relational database (RDBMS). Chit plans, users, bids, payments ellam foreign keys and ACID compliance oda secure-ah save aagum.
• **SQLAlchemy 2.0 (ORM):** Direct SQL queries ezhudha thevailla. Python classes/models moolama DB manage aagum. SQL Injection attacks 100% block pannum.
• **Alembic:** Database schema version control. Table structure changes-ah structured-ah upgrade/downgrade panna mudiyum.

4. External Services & Utilities

• **ReportLab:** Payment passbook, transaction receipts and statements dynamic PDF-ah generate panna.
• **OpenPyXL:** Admin monthly audit & balance sheets Excel (.xlsx) file-ah export panna.
• **qrcode + Pillow:** Dynamic UPI QR codes generate panna, so members easily scan panni pay pannalam.
• **SMTP Email:** Due date reminders and payment confirmation mail automatic-ah notify aaga.

5. Top Viva Questions & Instant Answers

Viva Question	How to Answer (Thanglish)
Why FastAPI over Django / Flask?	FastAPI async architecture (`asyncio`) moolama high concurrency support pannum. Built-in Pydantic data validation and auto Swagger `/docs` documentation iruku.
Why Vite instead of Create-React-App?	CRA uses Webpack which bundles slow. Vite browser-oda native ES Modules (ESM) use pandrathala instant dev server launch and fast hot reloading kedaikkum.
How is authentication secured?	Stateless JWT Tokens headers-la pass aagudhu. Passwords plain text-ah pogama cryptographic salt panni Bcrypt hashing vachu store pannirukkom.
How is Database integrity maintained?	SQLAlchemy ORM + RDBMS foreign keys use pannirukkom. Transactions commit/rollback aaguradhala payment/auction calculations-la partial write aagathu.